



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

In-Semester -01-Examination (Semester-I) 2025-26

M.Tech Structures
Behavior of Concrete Structures

Civil Engineering

PCC-SE502

Date: 13/10/2025 Time: 16:30-17:15

Max. Marks: 20

- Note:
- i. All questions are compulsory.
 - ii. Assume suitable data if necessary
 - iii. Use of non-programmable calculator is permitted.
 - iv. This paper contains 1 page.

Q.N.	Question	Marks	CO	BT
a	What are the different types of concrete mixes as per IS 456?	03	CO 1	BT1
b	Mention the exposure condition types defined in IS 456.	03	CO 1	BT1
c	A 16 mm diameter Fe500 bar is used as tension reinforcement in a beam with M25 grade concrete. Calculate the development length (L_d) required for the bar in tension.	03	CO 2	BT 2
Q.2	a Find the maximum superimposed load that a simply supported beam can carry of size 230x450mm is having span of 4.5m. The beam is reinforced with 3-16+2-12mm Φ . The beam is subjected to extreme environmental conditions.	04	CO 3	BT4
b	Design a reinforced concrete cantilever beam having span 3m and is subjected to uniformly distributed dead and live load of 15kN/m and 12kN/m respectively. Use M25 and Fe500 grade of concrete and steel respectively.	08	CO 4	BT4

=====All the Best=====



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

In-Semester -01- Examination (Semester-I) 2024-25

First Year M.Tech Structures.

Civil Engineering

Seismic Analysis and Design of Structures

PEC-SE514

Date: 16/10/2025 Time: 16:30-17:15

Max. Marks: 20

- Note:
- All questions are compulsory.
 - Assume suitable data if necessary
 - Use of non-programmable calculator is permitted.
 - This paper contains 1 page.

Q.N.	Question	Marks	CO	BT
Q.1	a Write a short note on soft storey failure.	04	CO 2	BT 1
	b Write a note on influence of soil stiffness.	04	CO 1	BT2
Q.2	a Explain in brief amelioration of torsion effect.	05	CO 2	BT2
	b What are the causes of earthquake?	03	CO 2	BT2
	c Write a short note on response reduction factor.	04	CO 2	BT1

===== All the Best =====



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

ISE-I (Semester-I) 2025-26

Class: First Year M. Tech.

Branch: All branches

Course: Open Elective: Operations Research

Course Code: OEC-803

Date: 17-Oct-2025

Time: 16:30 to 17:15

Max. Marks: 20

Q. No.	Questions	Marks	CO	BT
Q. 1)	Define systems engineering. Differentiate between systems approach and system analysis. How is it related to operations research?	07	1	1
Q. 2)	What is a model? Explain in brief role of systems models and state the different types of system models.	07	3	2
Q. 3)	Explain the concept of system with a neat diagram.	06	4	2

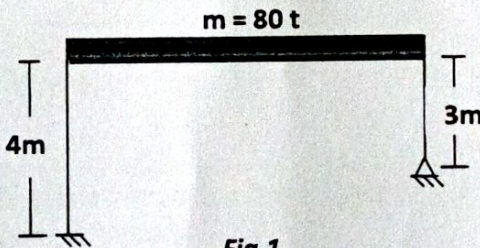
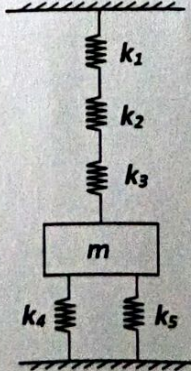
(Page 1 of 2)



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
In-Semester Examination I
Academic Year: 2025-26 (Semester – I)

Class: FY M.Tech. (Structural Engineering)	Subject: Structural Dynamics	Subject Code: PCC-SE503
Date: 14.10.2025	Time: 16:30 to 17:15	Max. Marks: 20

Note: i. Non programmable calculator is permitted.
 ii. Exchange/Sharing of stationery, calculator etc. not allowed.

Q.N.			Marks	CO	BT
Q.1	a	What are the essential characteristics of Simple Harmonic Motion (SHM)? Prove that an acceleration of a particle executing SHM is always directed towards the mean position and is proportional to its displacement.	05	CO1	BT1
Q.2	a	Derive a response of Single Degree of Freedom System for Damped free vibration.	04	CO3	BT3
Q.3	a	For the frame with rigid girder as shown in Fig.1; Compute the natural frequency and stiffness for horizontal motion. Take $I = 18823.5 \times 10^4 \text{ mm}^4$, $E = 2 \times 10^5 \text{ N/mm}^2$	05	CO2	BT2
 Fig.1					
Q.4	a	Consider the system as shown in Fig.2 having $k_1 = 2000 \text{ N/m}$; $k_2 = 1500 \text{ N/m}$; $k_3 = 3000 \text{ N/m}$; $k_4 = k_5 = 500 \text{ N/m}$. Find the mass if the system has a natural frequency of 10 Hz.	06	CO2	BT3
 Fig.2					

== ALL THE BEST ==



Class: FY M.Tech.

Course Name: Advanced Hydraulics

Course Code: PEC-WM505

Date: 15.10.2025

Time: 16:30 to 17:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	State and derive the law of conservation of mass and momentum	06	CO1	BT3
	b	A 4 m wide rectangular channel is carrying $10 \text{ m}^3/\text{s}$ at a depth of 2.5 m. There is a step rise of 0.2 m in the channel bottom. Assuming there are no losses at the transition, determine the flow depth downstream of the bottom step. Does the water surface rise or fall at the step?	04	CO2	BT3
Q.2	a	Explain the characteristic of boundary layer.	06	CO3	BT2
	b	What is surface roughness? Define (i) Isolated roughness flow; (ii) Wake interface flow; (iii) Quasi-smooth flow	04	CO3	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To identify and define fundamental concepts in hydraulics
CO 2	To analyse channel system performance and characteristics
CO 3	To study characteristics of different fluid flow

S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded. ISE-I Examination Sem-I AY 2025-26 Class : M.Tech		
Date: 15/10/2025 Time: 04:30-05:15 pm Exam duration : 45 min	Subject: Multimedia System Applications Course Code: PEC-AI506	Max Marks: 20

Note:-

- 1) Assume suitable data if necessary.
- 2) Use of calculator is permitted.

Q .No.	Questions	Marks	CO	BT
Q. 1	State and describe types of Multimedia. Explain the components of Multimedia. Give applications of Multimedia.	5	CO1	BT1, BT2
Q2	Explain in brief the following: What is meant by the terms Multimedia and Hypermedia? Distinguish between these two concepts.	5	CO1	BT1, BT2
Q4	What is an image? What are different image data type? Write in short about 8 bit colour image.	5	CO1	BT1, BT4
Q5.	Describe the different file formats used to store image and video data.	5	CO3, CO4	BT3, BT4

Blooms Taxonomy (BT):

BT1-Remember, BT2- Understand, BT3-Apply, BT4-Analyze, BT5-Evaluate, BT6-create

Course Outcome (Cos):

CO1- understand different types of multimedia data and basics of image and video.

CO2-Understood colour models of Image and video.

CO3- Analyze and design different compression algorithm.

CO3- Analyze and implement different compression standard for image.

CO4- Analyze and implement different compression standard for video.

CO5-Understand the transmission of multimedia data over communication network.

CO6-Understand Multimedia Database



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
In-Semester Examination I
Academic Year: 2025-26 (Semester – I)

Class: FY M.Tech.
(Structural Engg)

Subject: Bridge Analysis & Design

Subject Code: PEC – SE 506

Date: 15-10-2025

Time: 16:30 to 17:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Non-programmable calculator is permitted.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.
- v. Suitable data is given to you no need to carry IRC, any IS code during exam.

Q.N.		Question	Marks	CO	BT
Q	1	Classify bridges according to their structural form and explain each with examples.	05	CO2	BT2
Q	2	What is the significance of IRC Class AA, A, and B loadings?	05	CO3	BT1
Q	3	Explain the importance of site selection for bridge construction	05	CO2	BT2
Q	4	Illustrate with neat sketches five key bridge components and label them	05	CO2	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO	On successful completion of this course, students should be able to:
CO 1	To recall the fundamentals of water resources engineering.
CO 2	To understand the component of bridge, its classification and importance.
CO 3	To apply various theories of lateral load distribution for component design of bridge
CO 4	To analyse any given type of bridge.
CO 5	To get exposure to evaluation of sub structures, type of foundations, importance of bearing, lessons from bridge failures.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
ISE-I Examination (Semester - II)
Academic Year: 2025-26

Class: FY-M.Tech. (Common to All)

Subject: Project Management

Subject Code: AUD-901

Date: 13/10/2025

Time: 16:30 pm to 17:15 pm

Max. Marks: 20

Note:

- i. Non-programmable calculator is permitted.
- ii. Figures to right indicate full marks.
- iii. This paper contains two pages

Q.No.		Question	Marks	CO	BT																																	
Q 1	a	Define project management. Explain its principles and functions with examples.	05	CO1	BT1																																	
	b	Explain the importance of project management. State the Classification of Project.	06	CO1	BT2																																	
Q 2	a	Explain the concept of a Gantt chart with a simple example.	05	CO2	BT1																																	
	b	Define: 'Event & Activity' and 'Critical Path'. Draw a network for given activities of Project. The sequence of activities and their Predecessor activities is given in table below: <table><tr><th>Name of the Activity</th><th>Description of Activity</th><th>Predecessor Activity</th></tr><tr><td>A</td><td>Product design and specification finalization</td><td>--</td></tr><tr><td>B</td><td>Procurement of raw materials</td><td>--</td></tr><tr><td>C</td><td>Inspection and testing of raw materials</td><td>A</td></tr><tr><td>D</td><td>Machine setup and tooling preparation</td><td>B</td></tr><tr><td>E</td><td>Manufacturing / Production of components</td><td>A</td></tr><tr><td>F</td><td>Sub-assembly of components</td><td>C, D</td></tr><tr><td>G</td><td>Final assembly of product</td><td>E, F</td></tr><tr><td>H</td><td>Quality inspection and testing of final product</td><td>F</td></tr><tr><td>I</td><td>Packaging and labeling</td><td>G, H</td></tr><tr><td>J</td><td>Storage and dispatch to customer</td><td>I</td></tr></table>	Name of the Activity	Description of Activity	Predecessor Activity	A	Product design and specification finalization	--	B	Procurement of raw materials	--	C	Inspection and testing of raw materials	A	D	Machine setup and tooling preparation	B	E	Manufacturing / Production of components	A	F	Sub-assembly of components	C, D	G	Final assembly of product	E, F	H	Quality inspection and testing of final product	F	I	Packaging and labeling	G, H	J	Storage and dispatch to customer	I	05	CO2	BT2
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== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO	On successful completion of this course, students should be able to:
CO 1	Understand the concepts and functions of project management.
CO 2	Apply the project plan planning and monitoring techniques
CO 3	Analyze the project value, risk and quality.
CO 4	Design and develop projects at each stage of the software development life cycle (SDLC).



Class: FY M.Tech.

Course Name: Surface Water Hydrology

Course Code: PCC-WM502

Date: 13.10.2025

Time: 16:30 to 17:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
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- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Rain gauge station X did not function for a part of a month during which a storm occurred. The storm produced rainfalls of 84, 70 and 96 mm at three surrounding stations A, B and C respectively. The normal annual rainfalls at the stations X, A, B and C are respectively 770, 882, 736 and 944 mm. Estimate the missing storm rainfall at station X.	04	CO2	BT3
	b	Illustrate in detail measurement of rainfall using radar.	03	CO1	BT2
	c	Describe in detail orographic type of precipitation	03	CO1	BT2
Q.2	a	Estimate the daily evaporation from a large reservoir using the empirical equation of Horton, Meyer and Rohwer from the following data: (Refer Table 2.1) Water surface temperature = 28°C ; Air temperature = 30°C Atmospheric pressure = 766 mm of mercury; Relative humidity = 42%; Wind speed at 0.5 m above ground level = 7.5 m/s	06	CO3	BT3
	b	What is pan coefficient? What are the major drawbacks observed while measuring evaporation using pans?	04	CO3	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To train students in surface water hydrology as a prerequisite for water resource management
CO 2	To impart in-depth knowledge of hydrologic processes, data collection and management, and estimation of parameters.
CO 3	To develop students for application of concepts and methods of hydrology to field problems.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Civil Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination I

Class: FY M.Tech.

Course Name: Theory of Elasticity and Plasticity

Course Code: PEC-SE508

Date: 16.10.2025

Time: 04:30 pm to 05:15 pm

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones are strictly prohibited.
- Use of non-programmable calculators is allowed.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Statement	Marks	CO	BT
1 a	Define the following terms in the context of solid mechanics: i) Elasticity, ii) Plasticity	2	1	1
1 b	What are the key assumptions in the theory of elasticity.	3	2	2
1 c	Explain the difference between body forces and the surface forces and give example of each.	2	2	2
2 a	The following state of stress acts at a point in body. $\sigma_x = 120 \text{ MPa}$, $\tau_{xy} = \tau_{yx} = 60 \text{ MPa}$ $\sigma_y = 90 \text{ MPa}$, $\tau_{yz} = \tau_{zy} = 75 \text{ MPa}$ $\sigma_z = 70 \text{ MPa}$, $\tau_{zx} = \tau_{xz} = 55 \text{ MPa}$ If $E = 200 \text{ GPa}$ and $\mu = 0.3$, determine the equivalent strain present.	3	3	3
2 b	At a point in a three-dimensional stress system, the strain components are given as follows: $\epsilon_x = 0.0025$, $\gamma_{xy} = 0.0003$ $\epsilon_y = 0.0015$, $\gamma_{yz} = 0.0002$ $\epsilon_z = 0.0010$, $\gamma_{zx} = 0.0001$ Using Young's modulus (E) = 210 GPa and Poisson's ratio (μ) = 0.28, determine the principal stresses acting at the point using generalized Hooke's law.	5	3	3
2 c	Derive the relationship between Young's modulus (E), Shear modulus (G), and Poisson's ratio (μ). Using this relation, compute the missing parameter if $E = 210 \text{ GPa}$ and $\mu = 0.25$.	5	3	3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To remember the basics of Strength of Materials, Theory of Structures.
CO 2	To understand the behaviour of various structural elements subjected to lateral buckling.
CO 3	To analyse various structural elements using different mathematical and engineering problems.
CO 4	To apply the fundamentals of energy and mathematical methods to solve various structural elemental problems.
CO 5	To evaluate the performance of structural elements using energy and mathematical methods.